

3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

01:47

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- **Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Code:

```
import numpy as np

# Input matrices

print("Enter Array1:")

arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")

arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)

a=np.hstack((arr1,arr2))

print("Horizontal Stack:")
```

```
print(a)
```

```
# Perform vertical stacking (vstack)
```

```
b=np.vstack((arr1,arr2))
```

```
print("Vertical Stack:")
```

```
print(b)
```

Explorer

Average time
0.068 s
67.75 ms

Maximum time
0.072 s
72.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 68 ms Debug

Expected output

Actual output

Enter Array1:

Enter Array1:

Enter Array2:

Enter Array2:

Horizontal Stack:

Horizontal Stack:

Vertical Stack:

Vertical Stack:

Average time

0.068 s

67.75 ms

Maximum time

0.072 s

72.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 66 ms

✓ Test case 2 66 ms

Debug



Expected output

Actual output

Enter Array1:

Enter Array1:

-1 -2 -3

-1 -2 -3

-4 -5 -6

-4 -5 -6

-7 -8 -9

-7 -8 -9

Enter Array2:

Enter Array2:

10 11 12

10 11 12

13 14 15

13 14 15

16 17 18

16 17 18

Horizontal Stack:

Horizontal Stack:

[[-1 -2 -3 10 11 12]]

[[-1 -2 -3 10 11 12]]

[-4 -5 -6 13 14 15]

[-4 -5 -6 13 14 15]

[-7 -8 -9 16 17 18]]

[-7 -8 -9 16 17 18]]

Vertical Stack:

Vertical Stack:

[[-1 -2 -3]]

[[-1 -2 -3]]

[-4 -5 -6]

[-4 -5 -6]

[-7 -8 -9]

[-7 -8 -9]

[10 11 12]

[10 11 12]

[13 14 15]

[13 14 15]

[16 17 18]]

[16 17 18]]